

Pretentious Self-Parsing and Self-Evaluation of λ -Calculus

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Quantum Many Body Theory Group

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String “ $(\lambda. 0)$ ”

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Concrete and Abstract Syntax

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$\langle \text{term} \rangle ::= \langle \text{nat} \rangle$ (Variable)
| $(\lambda. \langle \text{term} \rangle)$ (Function)
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- 1 Represent each token
- 2 Represent each AST constructor
- 3 Parser `e_parse` from token list to AST
- 4 Evaluator `e_eval` from AST to λ -expression

Theory: Algebraic Data Types and Scott Encoding

- Algebraic data type (ADT, 조립식 타입): Sum, product, and recursive types

Constructor	Scott Encoding
$\lfloor \text{LPAR} \rfloor$	$= \lambda / r a v. l$
$\lfloor \text{RPAR} \rfloor$	$= \lambda / r a v. r$
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Case statement

$$t \text{ fv fl fa} = \begin{cases} \text{fv } n & (t = \text{Var}(n)) \\ \text{fl } t' & (t = \text{Lam}(t')) \\ \text{fa } t_1 \ t_2 & (t = \text{App}(t_1, t_2)) \end{cases}$$

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- Scott: Top-level v, l, a operate on the immediate subexpressions directly (Top-down)
- For parsing and evaluating, **Scott encoding** is more suitable!

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`( $\lambda n. \text{lookup } lst\ n$ )`

(Var: Lookup)

`( $\lambda t. \text{eval\_comb } t\ (\text{cons } lst\ (\text{Var } 0))$ )`

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- Implementation in GitHub repository!

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